PlayerNation Ratings Engine — Methodology

Player + Coach engines · Wyscout FIFA World Cup 2018 dataset · Arindam Bhadra

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Executive summary

This submission delivers two working rating engines and one methodology. The player engine produces an 8-dimension performance profile plus a role-aware composite for every player who appeared in FIFA World Cup 2018. The coach engine produces a 5-dimension profile plus an equal-weighted composite for every head coach at the tournament, deliberately measuring *observed coaching-attributable performance* rather than achievement.

Both engines share four commitments: only observable evidence enters a score; explanation is a first-class output, not an afterthought; descriptive style is separated from ability; and a rating is a summary of an explanation, not the product itself. Everything else — dimension counts, weight tables, shrinkage constants, confidence bands — is a defensible design choice, tunable as more data becomes available.

This methodology document covers the *why* — philosophy, tradeoffs, and evolution. The companion `README.md` in the repo covers the *how* — setup, run commands, folder structure, and column-level schema for all output files.

The product bet

A single number is a leaderboard artifact. It is legible and shareable, but it is not *actionable*. PlayerNation's mission — *help every player understand and improve their performance* — demands actionability, and actionability demands dimensions. A player who is told they are a 74 learns nothing; a player who is told their Ball Progression is elite but their Chance Creation is below role median learns exactly what to work on. The engines therefore commit to a **multi-dimensional profile plus one headline composite**: the composite for engagement, sharing, and comparison; the profile for understanding and coaching.

The presentation layer envisaged (not built in v1) is a player card with a radar chart of dimensions, a headline composite, and one signature-action callout — plus, for every score, an expandable "what drove this" view listing the top contributing action categories and their weights. Explainability is the product.

Shared philosophy (both engines)

Performance vs contribution

Performance is how well a player executed their own action. **Contribution** is how much that action helped or hurt the team. A perfectly executed safe back-pass has high performance and low contribution. A failed line-breaking pass has low performance and ambiguous contribution. The engines score contribution weighted by action quality, not raw counts.

Explain first, rate later

Every dimension score retains its top contributing actions. The rating summarises the explanation; it is not itself the explanation. This is enforced in the outputs — `player_explanations.csv` and `coach_explanations.csv` are shipped alongside the ratings, not as a debug artefact but as the primary product surface for coaching.

Only observable evidence

The engines evaluate what the event data can support: situation, action, and outcome. They do not model intent, luck, opponent tactical instructions, or hidden coaching thought. Where evidence is thin, the engines say so — dimensions with weak data support are cut, and confidence bands communicate small-sample uncertainty rather than hiding it.

Style is not ability

Behaviour (aggression, discipline, risk appetite, reliability) and coaching style (possession orientation, block height, attack tempo, discipline) are descriptive traits. They appear in the profile but never in the composite. Aggression is a style, not a virtue; the engine refuses to convert style into a moral judgement.

Field-relative, honestly

Percentile-within-role (players) and percentile-within-field (coaches) mean scores are relative to *this* dataset. A player's 78 means "top-quintile for their role at WC 2018," not "objectively good football." Absolute anchoring requires cross-tournament and cross-league data — a v2 unlock, named upfront, not hidden.

Player engine

What a player rating means

A player's **composite** is the mean percentile across the dimensions their role is *expected* to deliver, computed within their position group. The **profile** is the full 8-dimension view — always shown, always available for explanation, and capable of surfacing heroic moments outside role (a forward's goal-line clearance appears in the card even though it does not feed the composite). The composite is not strictly cross-position comparable, and that is a feature: a keeper's 78 and a striker's 78 each mean "strong for their role."

The eight dimensions

Dimension	What it captures
Ball Recovery	Winning the ball back — interceptions, defensive duels won, loose-ball recoveries.
Ball Retention	Keeping possession under contact — accurate passes, penalised for turnovers and dangerous losses.
Ball Progression	Moving the ball forward meaningfully — final-third entries, line-breaking passes, passes into the box, forward carries.
Chance Creation	Manufacturing shots for team-mates — assists, key passes, accurate crosses, passes into the box.
Finishing	Converting chances — goals, shots on and off target.
Defensive Disruption	Breaking up opposition attacks — clearances, sliding tackles won.
Goal Protection	Goalkeeping and last-line defending — saves, reflex saves, sweeper actions, own-box clearances.
Transition	Value in transition moments — counter-attack tagged actions, keeper forward launches. Deliberately thin in v1.

Role-aware composite

Averaging all eight dimensions dilutes the score on role-irrelevant work. A stress-test on A. Badri — a Forward who spent 224 minutes pressing high and rarely finishing — surfaced this: his goal-protection score of 26.8 dragged a composite that should have read as "hard-working link forward, not a goalscorer." The fix separates the **complete profile** (all 8 dimensions, always shown) from the **headline composite** (mean of the dimensions *expected* for the role):

Role	Composite inputs (expected dimensions)
Goalkeeper	Goal Protection · Retention · Transition
Defender	Recovery · Defensive Disruption · Retention · Progression
Midfielder	Retention · Progression · Recovery · Chance Creation · Transition
Forward	Finishing · Chance Creation · Progression · Retention

The role → expected map is a configurable product decision, not a hardcoded rule. As football understanding sharpens or new roles are surfaced in the data, the map is tunable in one place.

Scoring pipeline

event → quality-weighted contribution → per-90 rate
→ minutes-weighted shrinkage toward role mean
→ percentile within position group
→ dimension score (0–100)

Composite = mean of expected dimension scores per role.

Why each step is there

- **Per-90 normalisation.** A player with 700 minutes cannot outrank one with 200 by piling on volume. Rates, not totals.
- **Minutes-weighted shrinkage (K = 300).** Per-90 rates are noisy on small samples. Each player's rate is pulled toward the positional mean with weight `w = minutes / (minutes + K)`. K = 300 means roughly three full matches of evidence before a player's own rate outweighs the positional baseline — deliberately conservative for a 64-match tournament. K is a fairness-vs-responsiveness dial; higher K is more sceptical of small samples.
- **Percentile within position group.** Keepers scored against keepers, strikers against strikers. Cross-role fairness is built into the arithmetic, not bolted on afterwards. Nobody floors out for not doing another position's job.

Contribution weight tables (a defensible sample)

Weights are hand-set for explainability. A learned-weight model (xT, xG-style outcome models) is a v3 direction; v1 chooses transparency. A defender who sees their Progression score can trace it directly to `final_third_entry × count` plus `pass_into_box × count`, and see the same weights that produced anyone else's score.

Dimension	Sample weights (action → weight)
Recovery	interception 3.0 · def duel won 2.0 · loose ball won 1.0 · def duel lost −0.5
Retention	accurate pass 0.10 · lost ball −1.0 · dangerous lost −3.0
Progression	pass into box 4.0 · final-third entry 3.0 · line-breaking pass 3.0 · forward carry 1.5
Chance Creation	assist 8.0 · key pass 4.0 · pass into box 1.5 · accurate cross 1.0
Finishing	goal 8.0 · shot on target 2.0 · shot off target −0.3
Goal Protection	reflex save 5.0 · save 4.0 · keeper sweep 2.0 · own-box clearance 1.0

Weights are directional and tunable — every value here is a v1 choice made with awareness that it could be argued the other way. The point is that the choice is visible.

Confidence bands, not hard cutoffs

Rather than excluding low-minute players, every player is displayed with a confidence band: **High** at 450+ minutes, **Medium** at 180–450, **Low** below 180. Low-band scores are labelled *indicative*. This aligns with the brief's "improve as more data becomes available" and treats uncertainty as information rather than a reason to hide.

Behaviour layer

Four traits sit alongside the composite, each computed as a percentile within role and each purely descriptive:

- **Discipline** — inverse of fouls, cards, and simulations per 90.
- **Aggression** — duels, tackles, and fouls per 90. A descriptor, not a judgement.
- **Risk appetite** — share of ambitious passes out of all passes.
- **Reliability** — pass completion, penalised for dangerous losses.

Behaviour lives in the player engine (not the coach engine) because behaviour is an *individual* property. The coach engine will later *aggregate* it — squad-level discipline and composure after conceding are coaching signals — but it does not originate there.

Thirds as context

The pitch is split into defensive, middle, and attacking thirds to explain *where* a player's value came from. Thirds do not receive their own score, and in v1 they do not yet multiply action value. Contextual valuation — a lost ball in your own third costing more than one in the opponent's — is a locked v2 lever.

Worked example — A. Badri (Forward, 224 min, Medium confidence)

> Strengths: Retention 89, Recovery 68, Progression 68, Disruption 63. Weaknesses: Chance Creation 20, Transition 30, Finishing 38. Goal Protection 27 sits in the profile but is excluded from the Forward composite. Thirds distribution: 9% defensive / 44% middle / 46% attacking. The recovery score comes primarily from 9 interceptions, mostly in the *attacking* third (pressing from the front — 15 attacking, 7 middle, 6.5 defensive). The engine reads him as what he is: a hard-working, high-pressing link forward, not a goalscorer.

Player engine — known limitations

- **Transition is thin.** Real transition is sequence-based (recover-and-immediately-progress); v1 uses counter-attack-tagged actions plus GK forward launches only. Named, not hidden.
- **Defensive Disruption is thin.** The `Blocked` tag sits on the shooter's event, not the defender's — attribution is not clean. Blocks are not credited in v1.
- **Finishing merges** open-play, headers, penalties, and free-kick shots. Kane-style penalty inflation risk. Penalty-goal discount is a candidate quick fix; full shot-type split is v2.
- **Position groups are coarse.** Wyscout roles: GK / Def / Mid / Fwd. No fullback-vs-winger split in v1.
- **No age or level fairness.** The World Cup is elite-adult only. Cross-league and cross-age fairness needs v2 data.
- **No context modifiers.** Scoreline, minute, tournament stage, and game state do not yet affect action value. Locked v2 lever.

Coach engine

The load-bearing sentence

> *How well did this team perform in ways that can reasonably be attributed to coaching, based on observable match evidence?*

Every metric in the coach engine has to help answer that exact question. If it does not, it belongs in the style profile, the context layer, or gets cut. This wording is deliberate — three earlier drafts said "coach's influence" before I realised *influence* is a causal word that implies a delta from a baseline (the team with vs without this coach), which the data cannot provide. The engine therefore measures *level* (where this team's coaching-attributable performance ranks), not *delta* (how much the coach moved it). This is a more defensible claim, not a weaker one.

Quality, not achievement

The engine rates observed on-pitch behaviour, not outcomes or progression. France winning the World Cup with a deep block, low possession, and clinical finishing correctly ranks 9th on performance. Germany, playing well and going out at the group stage, ranks 3rd. Achievement lives in the context layer (deferred to v1.1), sibling to the rating, never a parent. If the champions had topped the performance table, the engine would be secretly rewarding results — the fact that they do not is the philosophy proving itself.

Attribution: the piece the player engine did not need

Not everything a team does is the coach's doing. The engine scores only coaching-attributable behaviour, and is explicit about the split. This is what makes the engine intellectually defensible — it reasons about *who gets credit*, not just *what happened*.

Evidence	Coach influence	Player influence
Build-up structure & progression	High	Medium
Defensive shape & compactness	High	Medium
Pressing coordination	High	Medium
Set-piece routines	High	Medium
Substitution timing & post-sub swing	High	Low
Chance creation volume	Medium	Medium
Discipline	Medium	Medium
Finishing quality	Low	High
1v1 dribbling	Low	High
Shot-stopping	Low	High
Penalty conversion	Low	High

Repeatable-pattern clause: player-executed actions contribute to coach ratings only when they reflect repeatable team patterns, not isolated individual execution. Twenty shots a match is a coaching signal; whether they went in is finishing.

Attribution is enforced in code, not just prose: open-play goals are excluded from Chance Creation (only the shot location is credited, so a central-box goal and a central-box miss score identically); only set-piece goals feed a coach dimension; penalties, throw-ins, and goal kicks are excluded from Set Pieces.

Why five dimensions, not eight

The first draft mirrored the player engine with eight dimensions. The data audit killed three of them: no substitution data in the event stream (a dedicated 64-match metadata scan confirmed no lineup-change fields), no formation tags, no tracking data for compactness or off-ball shape. Fewer dimensions with strong evidence beat more dimensions with weak proxies — and it matches the philosophy of measuring only what is observable.

Dimension	Evidence base
Build-up Quality	56k pass events with coordinates and tags — strong.
Chance Creation	1,419 shots with position and tags — strong.
Set Pieces	5,964 free-kick events with subtypes and sequence tracking — strong.
Defensive Organisation	Duels, fouls, clearances, opponent shots — a documented proxy.
Pressing Structure	Recovery and duel locations in the opposition half — a documented proxy, no PPDA or tracking.

Scoring pipeline

event → quality-weighted contribution → per-match rate
→ percentile within the 32-team field
→ dimension score (0–100)

Composite = equal-weighted mean of the five dimension scores.

Two intentional differences from the player engine: **per-match** (coaches do not come off, and matches are the natural denominator) rather than per-90; and **percentile within the 32-team field** rather than within role (no roles exist for coaches). No shrinkage in v1 — confidence bands carry the uncertainty in a form a coach can actually read.

Why equal weights across dimensions

There is no ground truth for coaches to justify unequal weights. Equal weighting is the honest default; any deviation would be an unverifiable claim. Within-dimension weights (how actions rank *inside* one dimension) are the same mechanism the player engine uses.

Why no separate thirds layer

The player engine exposes a `player_thirds.csv` and `player_thirds_detail.csv` because *where a player operates* is genuine additional signal about *who that player is* — a striker in the attacking third and a fullback in the defensive third read differently on the same pitch. The coach engine deliberately does not carry an equivalent layer, for two reasons.

First, the dimensions are already location-shaped by construction. Build-up Quality is a middle-to-attacking-third metric (final-third entries, line-breaking passes, passes into the box). Defensive Organisation is an own-half metric (own-half interceptions, own-third defensive duels, clearances from the box). Pressing Structure is an opposition-half metric (opposition-half interceptions, opposition-third duels won). Chance Creation lives in and around the box by definition. A `coach_thirds.csv` would restate the dimensions in another format, not surface new information.

Second, location *is* doing work in the coach engine — it just does it inside the weight tables rather than as a separate layer. An opposition-third interception scores higher than an opposition-half one (3.0 vs 2.0) inside Pressing; a clearance from the box carries a specific weight that a clearance elsewhere does not. Third-of-pitch is baked into the action taxonomy itself, which is the honest place for it in a coach model: location modifies value where the football justifies it, without pretending to be an independent dimension.

The tradeoff is transparency. A reviewer scanning the CSV list might reasonably ask *why do players get a thirds file and coaches do not?* This section is the answer: because for a coach, thirds are inside the dimensions, not alongside them.

Confidence bands (matches played)

High at 6+ matches (semi-finalists and finalists), **Medium** at 4–5 (round-of-16 exits), **Low** at 3 (group exits). A three-match rating is *indicative*, not stable. Do not over-read Germany at #3 — the band does that work.

Style layer (profile-only)

- **Possession orientation** — team passes ÷ all passes in the match.
- **Block height** — mean x-coordinate of defensive actions (interceptions, clearances, defensive duels).
- **Attack tempo** — passes per possession sequence; low = direct, high = patient.
- **Discipline** — fouls per match, inverted.

All four are percentiled and displayed. None enter the composite.

Worked examples — the philosophy in numbers

> **Brazil #1 (composite 90.6, Medium).** High possession, high creation, strong across the board (Pressing 100, Chance Creation 96.9). The right kind of team to top a *performance* table — the sanity check that the engine is not broken.
>
> **France #9 (composite 61.9, High).** Champions, mid-table on performance. Defensive Organisation 93.8 (elite, correct), Chance Creation 40.6 (below median — they did not create volume, they finished clinically, and finishing is excluded by design). Possession 44, block 28 — deep, pragmatic. The centrepiece: the champions did not top the coaching-performance table, and that is the point.
>
> **Germany #3 on 3 matches (Low).** Group-stage exit ranked third — played well, went out. A results-based system buries them; a quality-based one surfaces them. Confidence band flags the three-match instability, and the ranking becomes readable in that light.

Coach engine — known limitations

- **Defensive Organisation conflates volume with quality.** It is driven heavily by own-half interception *volume*, which rewards teams that simply defend a lot (camped deep, under pressure). France (deep elite block) scores correctly high, but Sweden 100 / Egypt 90.6 / Iran 87.5 are inflated by defending-a-lot rather than defending-well. The strongest v1.1 fix: normalise defensive actions per opponent possession.
- **Three-match ratings are indicative, not stable.** Half the field played only three games. Confidence bands communicate this; do not over-read a Low-confidence rank.
- **Pressing is a proxy.** No PPDA, no tracking. High recoveries and high duels won only — the standard event-data pressing proxies, but weaker than tracking-based ones.
- **Field-relative, not absolute.** A 70 means top-third-of-this-tournament, not objectively-good-coaching.
- **In-Game Management absent.** No substitution or formation data in the event stream. A whole coaching dimension the data cannot support.
- **No context layer in v1.** Squad strength, opponent quality, and progression are deferred to v1.1 — sibling to the rating, never in the composite.

Tradeoffs explicitly named

Decision	Chose	Alternative	Why
Rating shape	Composite + profile	Single number	Actionability demands dimensions; engagement demands a headline.
Cross-role fairness (players)	Percentile within role	Raw scores compared across roles	Fairness built into arithmetic; no role floors out.
Small-sample handling (players)	Shrinkage toward role mean	Hard minutes cutoff	Every player shown; uncertainty is information, not exclusion.
Small-sample handling (coaches)	Confidence bands only	Bayesian shrinkage	Coaches need to read the number; explainability wins over statistical precision.
Action weights	Hand-set, transparent	Learned via outcome models	Explainability first; learned weights are v3.
Coach baseline	Peer comparison within field	Expected performance given squad strength	"Coaching quality" is the question. "Compared to expectation" is achievement — different product.
Coach outcomes	Excluded from composite	Weighted into rating	Outcomes are supporting evidence, not primary. Prevents secret result-rewarding.
Coach xG-style modelling	Rejected	Adopt xG for chance creation	The player engine does not use xG. Consistency across engines.
Style	Profile-only	Weighted into rating	Style is not ability. Aggression is not a virtue.

Evolution path

v1.1 (near-term)

- **Defensive Organisation normalised per opponent possession.** Removes the volume-from-being-pinned inflation. Priority fix.
- **Context layer for coaches.** Squad strength (FIFA-ranking prior), opponent quality faced, progression — exposed as siblings to the rating on the coach card, enabling the

"exceeded/met/underperformed expectation" narrative without contaminating the quality score.

- **Penalty-goal discount** in player Finishing. Full shot-type split still v2.
- **Coach mapping automation** — derive from raw match metadata coach IDs instead of the static file.
- **Substitution investigation** — check whether the raw `matches.zip` surfaces usable in-game-management signal.

v2 (data unlocks)

- **Contextual action valuation.** Same action, different value by third and game state. A lost ball in your own third should cost more than one in the attacking third.
- **Cross-league data.** Age fairness, style fairness, larger sample stability, absolute anchoring (a fixed "good football" reference).
- **Lineup and tactical data.** Real coach signal — formation heatmaps, sub-timing impact, in-game adaptation.
- **Richer Transition and Pressing.** Sequence-based (recover-and-immediately-progress), pressing triggers — if tracking data ever arrives.
- **Aggregated player behaviour as a coach signal.** Squad-level discipline and composure-after-conceding, sourced from the player engine's behaviour layer.
- **Player behaviour trends and volatility** across matches — from a static profile to a trajectory.

v3 (product-level extensions)

- **Learned action weights** via outcome models (xT, xG frameworks) — with an explanation surface that still lets a player see *why*.
- **Individual improvement tracking.** Rating deltas over time as coaching feedback.
- **Peer-group comparison at grassroots.** Same age, position, level — for actual PlayerNation users, not just pros.
- **Player-role recommendation and team-fit engine.** Compatibility, pairings, squad balance.
- **Development recommendations.** Turn insights into actionable coaching prompts ("reduce risky passes when building out from your own third").

Assumptions worth flagging

- **The Wyscout event data is a faithful representation of the match.** A raw-vs-processed audit confirmed identical event counts and taxonomy for the WC subset.
- **Position groups (GK / Def / Mid / Fwd) are meaningful units for percentile comparison.** A finer-grained role model (fullback vs winger, No. 6 vs No. 8) would improve fairness but requires role annotations the dataset does not carry.

- **The hand-set weight tables are directional and open to revision.** They are not optimised against an outcome model in v1. Any weight can be argued; the point is that the argument is visible.
- **Percentile-within-field is the right normalisation for a single tournament.** With cross-tournament data, absolute anchoring becomes possible and preferable.
- **Substitution data is out of scope for coach evaluation in v1.** The processed WC event files do not surface it as coach-attributable evidence; the raw `matches.zip` does contain substitution records, and a v1.1 investigation is planned.

Deliverables

Working solution. Two single-file Python engines — `player_engine.py` and `coach_engine.py` — each reading the same processed Wyscout dataset. Player engine consumes the processed event JSON files plus the raw `matches.zip` (for real minute reconstruction from substitution records). Coach engine consumes the event JSON files plus a static `coach_mapping.json` (32 head coaches, Wikipedia-sourced).

Outputs. Nine CSVs in `output/`:

Player engine (5 files):

- `player_ratings.csv` — rank, id, name, role, minutes, confidence, composite, 8 dimension scores.
- `player_behaviour.csv` — 4 behaviour traits (percentile within role).
- `player_explanations.csv` — top action drivers per player per dimension. The primary explainability surface.
- `player_thirds.csv` — share of positive value by pitch third (defensive / middle / attacking). Answers *where* a player operated.
- `player_thirds_detail.csv` — contribution per dimension per third. Answers *what* they did in each area of the pitch.

Coach engine (4 files):

- `coach_ratings.csv` — rank, team_id, coach, team, matches, confidence, composite, 5 dimension scores.
- `coach_style.csv` — 4 style descriptors (profile-only).
- `coach_explanations.csv` — top action drivers per coach per dimension.
- `coach_matches.csv` — raw per-match contribution per coach per dimension. Enables match-level audit of any tournament rating.

Run. `python src/player_engine.py` and `python src/coach_engine.py` from project root. Full run instructions, dependency setup, and column-level output schema are in `README.md`.

Closing note

The most useful thing this exercise surfaced was the discipline of separating what the data can support from what football knowledge suggests *ought* to be true. Every dimension that was cut, every metric moved from composite to style, every rating that surprised (France #9, Germany #3) was a small vote for the philosophy over the intuition. A ratings system is only as trustworthy as the honesty of its boundaries; this document is an attempt to make those boundaries visible.